3D HOLOGRAM DESK BUDDY V2.0: INTERACTIVE SMART COMPANION

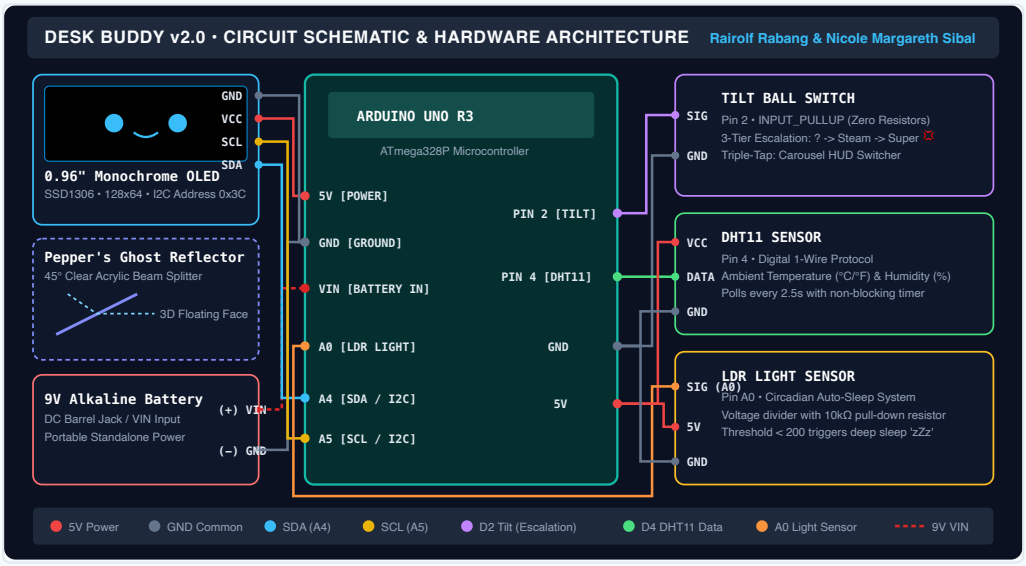
Final Case Study & Engineering Report

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1 Updated Image of Circuit Design (WOKWI / Tinkercad / Diagram)

Circuit schematic and Pepper's Ghost optical assembly of the 3D Hologram Desk Buddy v2.0. Powered by an Arduino UNO R3 from the MakerLAB Upgraded Learning Kit v2, the device uses a 0.96" I2C OLED display (SSD1306) projecting upward/downward onto a 45-degree clear acrylic reflector, alongside an interactive Tilt Ball Switch (Pin 2), Photoresistor LDR (Pin A0), and DHT11 Temp/Humidity module (Pin 4) compactly organized on an extra-small mini breadboard.

Simulation Link: [https://wokwi.com/ (Optional Wokwi or Tinkercad simulation link)](https://wokwi.com/)



2 Picture/s of the Finished Project

Photographs of the physical Desk Buddy prototype showcasing the clear optical reflector prism, mini-breadboard layout, 9V battery compartment, creator splash screen, and active floating hologram displays (companion faces, 3-tier anger escalation, climate dashboard, digital clock, and daily quotes).

📸 *[Finished Project Photo Placeholder]*
Place photo at `assets/finished_project_1.jpg` to embed automatically.

3 List of Materials / Components / Wiring Config

A. Bill of Materials

Component / Item	Quantity	Specification / Role
Arduino UNO R3	1 pc	MakerLAB Upgraded Starter Kit ATmega328P microcontroller board
0.96" Monochrome I2C OLED Display	1 pc	128x64 pixels (SSD1306 driver) - 3D hologram projection source
Tilt Ball Switch Module	1 pc	Motion sensor on Pin 2 triggering interactive 3-tier progressive anger escalation (? -> Steam -> Super Angry 💢) and HUD menu navigation
Photoresistor (LDR) & 10kΩ Resistor	1 set	Ambient light sensor on Pin A0 triggering circadian auto-sleep and wake
DHT11 Temp & Humidity Sensor	1 pc	Digital temperature & relative humidity sensor on Pin 4
Extra-Small Mini Breadboard (170 Tie-Points)	1 pc	Compact, portable, and aesthetic prototyping platform
Clear Acrylic / Transparent Plastic Reflector	1 pc	45-degree Pepper's Ghost optical beam reflector sheet
9V Battery Snap with 2.1mm DC Barrel Jack	1 pc	Regulated portable power input to Arduino DC jack

B. Wiring Configuration (I2C & Power)

Module / Source Terminal	Arduino UNO Pin	Connection Details
OLED GND	`GND`	Common system ground
OLED VCC	`5V`	5V power input from Arduino regulator
OLED SCL	`A5 (SCL)`	I2C Clock line
OLED SDA	`A4 (SDA)`	I2C Data line
Tilt Switch	`Pin 2 & GND`	Internal INPUT_PULLUP, zero external resistors
LDR Light Sensor	`Pin A0`	Voltage divider between 5V, Pin A0, and 10kΩ to GND
DHT11 Data	`Pin 4`	Digital temperature & humidity signal (VCC to 5V, GND to GND)

Module / Source Terminal	Arduino UNO Pin	Connection Details
9V Battery	`DC Jack / VIN`	Portable battery power supply with reverse-polarity protection

4 Estimated Cost of the Project Components Used

Component Item	Unit Cost	Total Cost
Arduino UNO R3 (MakerLAB Kit)	$6.00 - $10.00	**$7.50**
0.96" I2C OLED Display (SSD1306)	$2.50	**$2.50**
Tilt Ball Switch Sensor	$0.50	**$0.50**
Photoresistor (LDR) + 10k Resistor	$0.30	**$0.30**
DHT11 Sensor Module	$1.20	**$1.20**
Extra-Small Mini Breadboard & Jumpers	$1.50	**$1.50**
Transparent Acrylic / Plastic Hologram Plate	$1.00	**$1.00**
9V Battery & DC Barrel Snap	$2.00	**$2.00**

💰 **Overall Estimated Project Cost: $16.50 (Approx. ₱900 - ₱980)**

5 Full Working Code

Firmware sketch targeting ATmega328P with SSD1306 OLED driver:

```
/*
 * ========================================================================
 *  DESK BUDDY v2.0 - XIAOZHI INTERACTIVE FACE ENGINE & TRIPLE-TAP HUD
 * ========================================================================
 *  Microcontroller: Arduino UNO R3 (MakerLAB Upgraded Learning Kit v2)
 *  Display:         0.96" I2C Monochrome OLED (SSD1306, 128x64)
 *  Optics:          45° Transparent Pepper's Ghost Hologram Reflector
 *  Sensors:         - Tilt Ball Switch (Pin 2: Tap Combo Escalation & Triple-Tap)
 *                   - Photoresistor LDR (Pin A0: Circadian Auto-Sleep)
 *                   - DHT11 Sensor (Pin 4: Live Temp & Humidity)
 *  Form Factor:     Extra-Small Mini Breadboard (170 Tie-Points)
 *  Power:           9V Battery / USB 5V
 *
 *  XIAOZHI FACE ENGINE FEATURES (Adapted for Arduino Uno SSD1306):
 *    1. FLUID IDLE GAZE ENGINE: Liquid smooth Lerp gliding across the display!
 *       The eyes actively inspect their surroundings (glance left -13px, glance
 *       right +13px, look up curiously -7px, diagonal glances, and modest downward rest)
 *       with organic hold durations (1.5-3.0s) and seamless return-to-center.
 *    2. PROGRESSIVE SQUASH-AND-STRETCH BLINKING: Smooth frame-by-frame eyelid
 *       interp curve with natural blinks coordinated during wide gaze transitions.
```

```
 *    3. 3D PUPIL PARALLAX & GLOSS: Black pupils lead the gaze with an adorable
 *       white micro-reflection highlight dot for that glossy holographic shine.
 *    4. CONNECTED 3D MOUTH PERSPECTIVE: The mouth smoothly tracks the gaze direction
 *       with dampened parallax, making the entire face feel like a real turning head!
 *    5. 3-TIER ANNOYANCE ESCALATION (Single Tap):
 *       - Tier 1: Curious Question Mark (?) emote (starts 5s combo window)
 *       - Tier 2 (tap within 5s): Little Angry Face (\ / glare + pout + steam)
 *       - Tier 3 (tap within 5s): Super Angry Face (rage tremor, roaring mouth, throbbing 💢 v
 *       - Auto calm-down: Returns to calm mode if untouched for 5 seconds!
```

6 Short Video Test

Demonstration video showcasing the Desk Buddy hardware startup, OLED facial expressions, and battery-powered portable operation.

Video Link: ▶ Watch Demonstration Video (https://youtu.be/your_video_id_here)

Notes: Demonstration highlights the Pepper's Ghost 3D floating illusion, the interactive 3-tier progressive anger escalation (curious ?, huffing with billowing steam, super angry 💢 with rage tremors), triple-tap HUD menu switching, room light circadian auto-sleep reaction (zZz), DHT11 live climate dashboard, real-time clock, daily quotes, and creator startup splash screen.

7 Short Description of the Project Functionality

> The 3D Hologram Desk Buddy v2.0 is an interactive desktop companion engineered by Rairolf Rabang and Nicole Margareth Sibal using an Arduino UNO R3 from the MakerLAB Upgraded Learning Kit v2 and an extra-small mini breadboard for a sleek, compact, and aesthetic form factor. By utilizing a 0.96-inch monochrome I2C OLED display (SSD1306) and the optical Pepper's Ghost principle with a 45-degree transparent reflection plate, high-contrast graphics appear to float seamlessly in mid-air with zero visible display frame. The system features: (1) expressive animated companion faces with physical motion reactivity, including an interactive 3-tier progressive anger escalation engine (Single tap: Curious '?'; Second tap within 5s: Little Angry with alive huffing and billowing steam; Third tap within 5s: Super Angry with rage tremors and pulsating anger vein 💢, followed by a calm-down exhale) and triple-tap HUD page switching; (2) a circadian auto-sleep engine that detects room lighting via a photoresistor on Pin A0, gracefully putting the companion into deep sleep with floating 'zZz' holograms when lights are turned off; (3) a live DHT11 environmental climate dashboard reporting ambient Temperature (°C and °F), Relative Humidity (%), and Comfort index; (4) an automatic Real-Time Clock synced with computer compile time; (5) a rotating Quote of the Day stored in Flash PROGMEM; and (6) a custom startup splash screen crediting creators Rairolf Rabang & Nicole Margareth Sibal. Powered portably by a 9V battery with non-blocking timing, the Desk Buddy delivers an entertaining, functional, and futuristic companion for any workspace.